Czochralski crystal growing system CGS1218

Advanced silicon puller for the high demands of the semiconductor industry



The system

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The CGS1218 is designed for the semiconductor industry and enables the growth of high-purity silicon crystals with reliable reproducibility. Available with either shaft or cable head drive, the system offers flexibility for various process requirements. It provides precise, highly linear control of pull speed and rotation, supporting stable crystal growth and consistent quality. Its five-segment process chamber is suitable for a 32“ or 36“ hot zone, meeting both current and future production needs. Advanced automation and intuitive controls help reduce operator intervention and maximize productivity. The robust, modular design allows reliable, continuous operation and easy maintenance. The CGS1218 supports efficient and reproducible silicon crystal manufacturing for current and future demands.

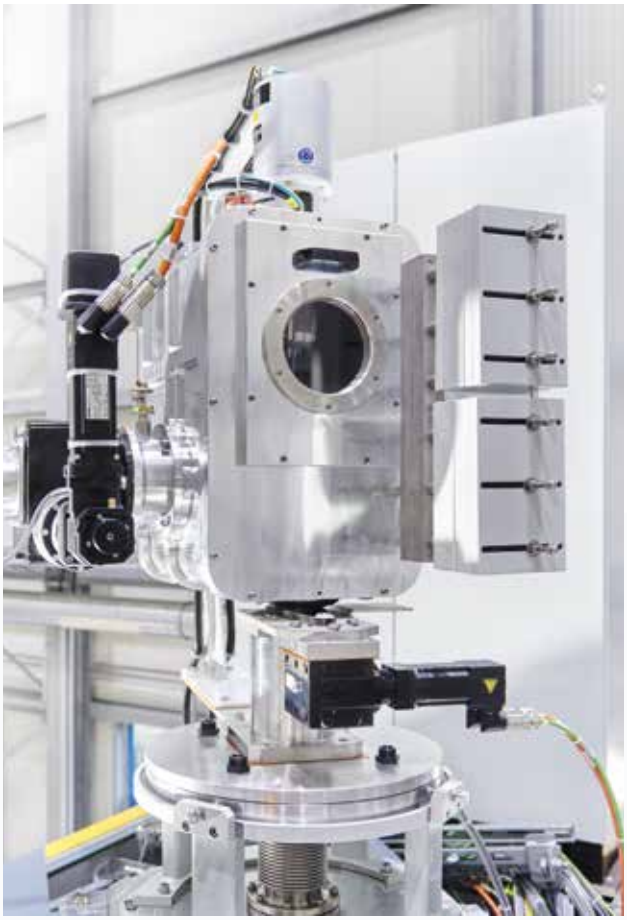
Advantages

- **Modular chamber design and system automation**
  » Less manpower, reduced human impact, higher efficiency
- **Shaft-type crystal drive unit (optional)**
  » High precision speed control, uniform results, excellent crystal properties
- **Superior machine and process stability**
  » Optimized fine-tuning, easy upscaling
- **Advanced safety concept for a modern work place**
  » Secures investment and ensures highest operational work safety.



The technology

The Czochralski process (Cz) is also known as “crystal pulling” or “pulling from the melt”. In this process, silicon (Si) is first melted and then allowed to freeze into a crystalline state in a controlled manner. The advantage of this method is that it is fast and highly controllable. The industrial production of high-purity monocrystalline silicon crystals using the Cz process has become well established primarily for the solar and semiconductor industries (in the computer industry for integrated circuits and in microsystem technology). You benefit from the high effectiveness and quality of the grown crystals.

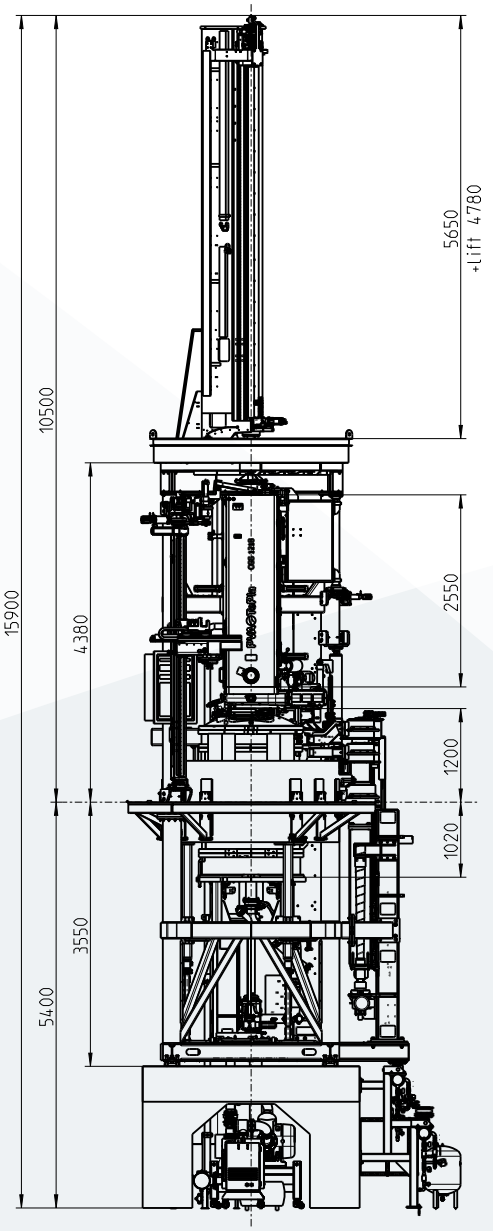


The process

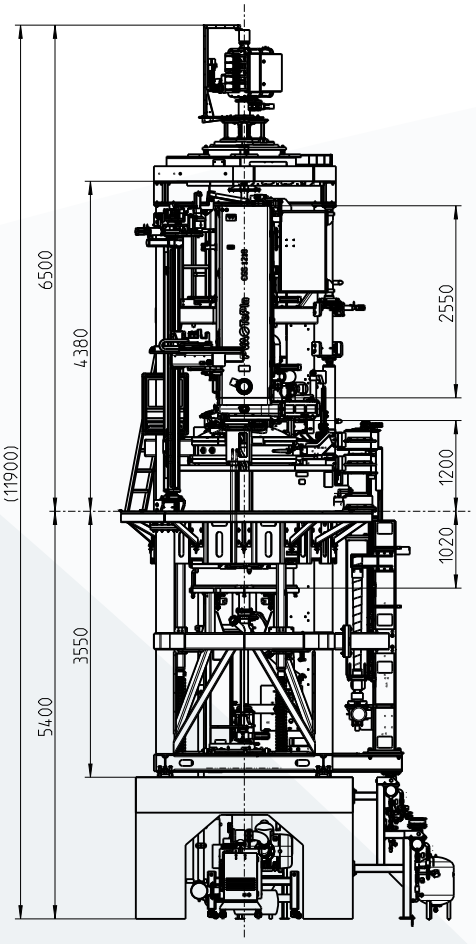
To begin with, high-purity polycrystalline silicon is placed into the silica crucible of a single crystal pulling system and then melted in a controlled atmosphere (argon) using a resistance heater. Once the temperature of the melt has stabilized (the melting point is around 1,412°C), a rotating monocrystalline silicon seed crystal is dipped into the melt. A slight temperature drop initiates the crystallization of silicon on the seed crystal. When the seed crystal is slowly pulled upward, a cylindrical silicon monocrystal hanging on the seed crystal starts to form. The pulling rate and temperature are regulated in such a way that a silicon monocrystal — whose orientation and structure is identical to those of the seed crystal—can be pulled with a constant diameter.

Dimensions (approx.)

Total height	» 14,050 mm (10,050 mm cable version) + pedestal
Height above operator level	» 10,500 mm (6,500 mm cable version)
Clearance height on lower level	» At least 3,550 mm above pedestal
Width	» approx. 4,100 mm
Depth	» approx. 3,300 mm
Required floor area (WxD)	» approx. 5,500 mm x 6,500 mm



shaft version



cable version

The system

Optional magnet

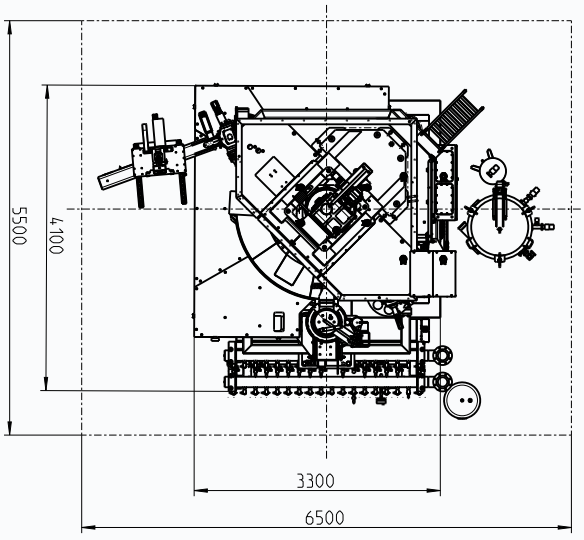
Magnet type	» Transversal or CUSP (up to 0.4 T)
Magnet, max. permissible weight	» Up to 16,000 kg
Magnet, max. permissible dimensions	» Outer Ø ≤ 2,050 mm, inner Ø ≥ 1,450 mm

Optional equipment

Crucible charging cart	» Available
Central feeder as recharging unit	» Available
Crystal removal device incl. cart	» Available

Optional services

Advanced process support	» Available
Advanced technology training, crystal growth	» Available
Remote access / support via VPN	» Available
Basic training of operators & service technicians	» Available



top view

Configuration

Handling systems, crucible and hot zone	»	Included
Cleanroom partition at crystal drive unit	»	Available
SiO filter system	»	Available
Vacuum pumping unit	»	Available
Heater pyrometer	»	Available
Safety unit for active crystal cooling	»	Available



Data sheet

Puller data	**Receiving chamber type**	Door type
	Weight	Approx. 23,000 kg w/o magnet
	Isolation valve, inner diameter	560 mm
	Crystal, diameter	Up to 310 mm for 12“ wafers, max. 450 mm
	Crystal, removable length	Up to approx. 2,800 mm
	Charge capacity	Up to 450 kg
	Hot zone, process chambers	Five segments with an inner Ø 1,250 mm
	Hot zone, process chambers, inner height	Approx. 2,120 mm
	Hot zone, crucible diameter	32“, max. 36“
Facility requirements	**Main line, nominal voltage** [1]	3 AC/PE 400 V ±5%
	Main line, nominal frequency [1]	50 Hz ±1%
	Main line, connected load [2]	Approx. 280 kVA
	Cooling water, total flow [2]	Approx. 600 l/min
Crucible drive unit	**Crucible drive, type**	Shaft
	Crucible drive, lifting speed [3]	0.01 - 260 mm/min
	Crucible drive, lift stroke [4]	Up to 900 mm
	Crucible drive, load capacity	Up to 700 kg
	Crucible drive, rotation speed [3]	0.1 - 20 rpm
Crystal drive unit	**Crystal drive, type**	Shaft or cable rotation head
	Crystal drive, pulling speed [3]	0.05 - 1,000 mm/min shaft version (0.05 - 560 mm/min cable head)
	Crystal drive, rotation speed [3]	0.1 - 20 rpm
Configuration	**PVA hot zone set**	It consists of 32" crucible and graphite parts, active crystal cooling device, tool set, stereo camera system and corresponding software. The hot zone is only available in combination with a separate hot zone license.
	Heater power supply with AFE technology	Included
	Concrete base / pedestal	Excluded
	Handling systems, process chambers	Combination of manual and electro motor-assisted swiveling and clamping systems
Magnet lift	**Magnet lift, max. stroke**	Included / 900 mm

1) Depends on actual configuration 2) Others on request 3) Stepless / Continuously adjustable 4) Actual stroke depends on hot zone

Our promise

We provide cutting-edge equipment and process technology for complex and highly sophisticated materials and material combinations that shift the boundaries of technological capabilities. The result: future-ready high-tech systems featuring industry-leading process technology and premium quality – Made in Germany.

From initial consultation to reliable after-sales support, we accompany our customers through every stage of the project with **personalized, expert service**. Our clients appreciate our flexibility – backed by the financial strength of **PVA TePla AG**, a publicly listed company.

This is how we consistently fulfill our promise: the best system technology for even better materials..

PVA Crystal Growing Systems GmbH
Im Westpark 10–12
35435 Wettenberg, Germany
Tel.: +49 (0) 641 68690 0
cgsinfo@pvatepla.com
www.pvatepla-cgs.com

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